

Extending Fitts' Law for Tapping on Surfaces Angled in the Depth Direction

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ABSTRACT

Recently, various shape-changing devices have been reported in the human-computer interaction field. However, the applicability of Fitts' law to these devices has not been efficiently discussed. Thus, in this study, we discuss the realization of a model that can predict the tapping time of displays angled in the depth direction. We first measured the tapping time of angled touch screens for 18 participants. Based on the results, we expanded and validated the Fitts' law model by considering the operator's line of sight. The results of regression analysis between the tapping time and the expanded model indicated that $R^2 = 0.97$ and $R^2 = 0.95$ for the displays angled to the front and back, respectively. We also considered the obstruction due to display height in the convex configuration and tapping difficulty in the concave configuration. The regression analysis for the modified model demonstrated $R^2 = 0.93$ for both front-angled and back-angled displays.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI); HCI theory, concepts and models.*

KEYWORDS

Fitts' law, Angled surface, Touch screen, Shape-changing display, Finger input

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1 INTRODUCTION

Recently, in the field of human-computer interaction, shape-changing devices that are composed of multiple display panels and can change their angles have been investigated and reported [16]. Commercial devices, such as foldable smartphones, are also available. Although these devices can represent various shapes, their operability may differ from that of a horizontal display surface. Therefore, a user interface for shape-changing devices that considers operability is important. To create a user interface with operability, the time for pointing tasks, such as selecting icons on the display using the mouse cursor or tapping with the fingertip, should be predicted. Fitts' law [11] is a model for predicting the time for pointing tasks. This model uses the target width and the distance from the starting point to the center of the target for prediction. In the case of tapping on a touch screen, there is a model that can reportedly predict the tapping time using the error between the target and tap position of the fingertip, as well as the standard deviation [4]. However, these models do not consider the environment comprising multiple-angled display panels. Therefore, in this study, we investigate the tapping time on touch screens angled in the depth direction. Furthermore, we expand the Fitts' law model for an environment of multiple-angled display panels.

2 RELATED WORK

2.1 Shape-changing Devices

In the studies on shape-changing devices, several configurations of multiple display panels have been reported. For example, Tilt Displays [1] represent the shape of content using nine displays that angle horizontally and vertically according to the content displayed. Paddle [17] is a multipanel shape-changing display that improves comprehension of content. Its shape can be changed to either a box or a circle by combining multiple panels. FoldMe [9] comprises two or three panels. The shape of this device can be changed by folding it like a book or angling one of the panels. Multi-plié [15] comprises multiple displays connected like an accordion and uses the shape change as information. Saniee-Monfared et al. [18] investigated an interaction design space for a convex configuration of a foldable device. Considering the shapes of these devices, we determined the conditions for the experiments in this study.

2.2 Modeling the Tapping Time on the Pointing Task

Fitts' law [11] is a model that can predict the pointing time T using the target width W and the distance from the starting point to the center of the target A . The model is expressed as follows:

$$T = a + b \log_2 \left(\frac{A}{W} + 1 \right), \quad (1)$$

where a and b denote the regression coefficients, and $\log_2 \left(\frac{A}{W} + 1 \right)$ is the task difficulty ID . This law was experimented on the 1D and 2D tasks [11, 12], indicating that it had high prediction accuracy on both tasks.

Numerous studies have explored the application and extension of Fitts' law to various types of targets and interaction scenarios, including expanding targets [13], small targets [7], and targets located at screen edges [2]. Extensions of Fitts' law to different interaction methods, such as 3D pointing tasks [6, 14] and gesture-based interactions [5], have also been investigated. Fitts' law has been examined in various display environments as well. FFitts law [4] was proposed as a model to predict tapping time while considering tapping precision on touchscreens. Bi et al. [3] explored how the internal bezel of large tiled displays affected user performance. Additionally, Hutchings [8] suggested that Fitts' law might underestimate the difficulty of movement tasks in multiple display environments.

Although some findings from these studies can be applied to shape-changing devices with multiple display panels, the effects of surface angles are not adequately considered. Therefore, this study specifically focuses on environments that involve multiple-angled display panels and extends the Fitts' law model to account for these factors.

3 INVESTIGATION OF TAPPING TIME ON DISPLAYS ANGLED IN THE DEPTH DIRECTION

3.1 Experiment Setting

An experiment was conducted to investigate the tapping time required on touch screens angled in the depth direction and to obtain data for the model expansion. In this experiment, two 7-in touch panels (Uctronics, U6104) were placed on a special pedestal to reproduce a display angled in the depth direction (Figure 1). The pedestal could be replaced with another pedestal to change the display angles corresponding to the experimental conditions.

The experiment was conducted on 18 participants (all male, mean age = 22.9, standard deviation = 0.94, all right-handed). The participants sat on a chair such that their body distance from the display was 250 mm, and their elbow heights were equal to the display surface height. The experiment was performed based on the 2D task of FFitts law [4]. In the 2D task (Figure 2), the start and target were depicted as circles, with the target width (W) corresponding to the diameter. The start and target locations in the 2D task were categorized into three types based on Fitts' law [12]: straight, 45° right oblique, and 45° left oblique. The participants conducted 16 trials of tapping the white start circle and the red target circle, in that order, with their index fingers. The tapping time for each

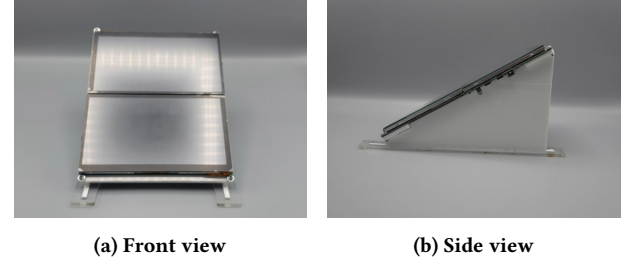


Figure 1: Experiment environment.

Table 1: Conditions for Same-angled Displays

Width W (mm)	Distance A (mm)	Angle ($^\circ$)
10	60	0, 15, 30, 45
15	120	-15, -30

trial was measured, and this procedure was performed for each experimental condition. The participants were instructed to tap the start circle and then the target circle as accurately and quickly as possible.

3.2 Conditions

The experiment was conducted using the same-angled displays presented in Table 1. The target width W and the distance A were set to two conditions, and the angle of the display to the horizontal was set to six conditions. The experiment was conducted under a total of 24 conditions combining these conditions. In Figure 3, the cases where the display was angled to the front and the back were defined as the positive and negative angles, respectively.

3.3 Results and Discussion

Figure 4 presents the average tapping time for the 18 participants. The error bars in the figure represent the standard deviations. In cases where the display was angled to the front, no apparent difference was observed in the tapping time depending on the angles (Figure 4a). Repeated measures ANOVA for the tapping time also revealed no significant difference. For the displays angled to the back, the tapping time appeared to slightly increase as the angle increased (Figure 4b). Repeated measures ANOVA, followed by Tukey's multiple comparison, revealed a marginal difference ($p < 0.1$) between the conditions of 0° and +30° in the case of $A = 60$ and $W = 10$. Therefore, compared with the tapping time at 0°, the tapping time tends to increase as the angle increases.

A regression analysis was conducted using ID in Fitts' law [11] as an explanatory variable to verify whether ID can predict the tapping time on displays angled in the depth direction. Figure 5 presents the results. For displays angled to the front, adjusted $R^2 = 0.97$, indicating that the tapping time is sufficiently predictable. However, for the displays angled to the back, adjusted $R^2 = 0.82$, indicating that the tapping time predictability is insufficient. Consequently, we hypothesized that the width of the target when looking at the display from the operator's line of sight would affect the tapping time.

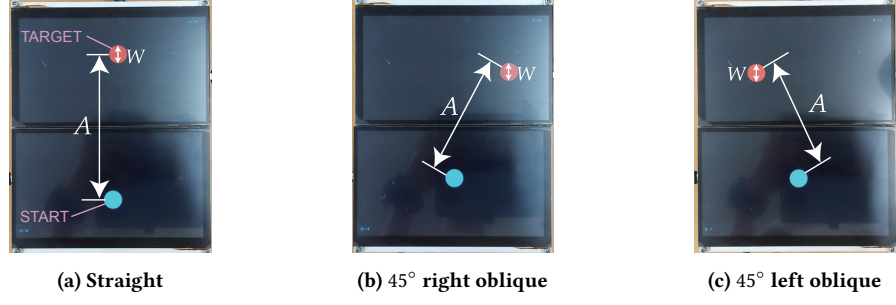


Figure 2: Start and target positions.

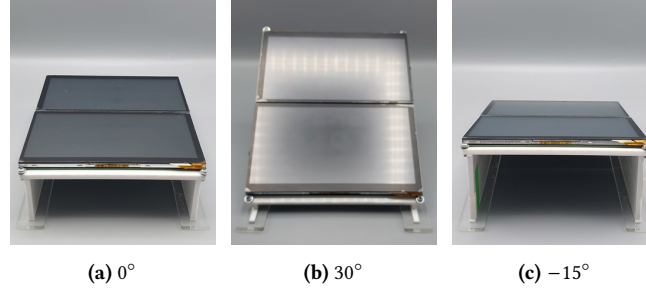


Figure 3: A set of same-angled displays.

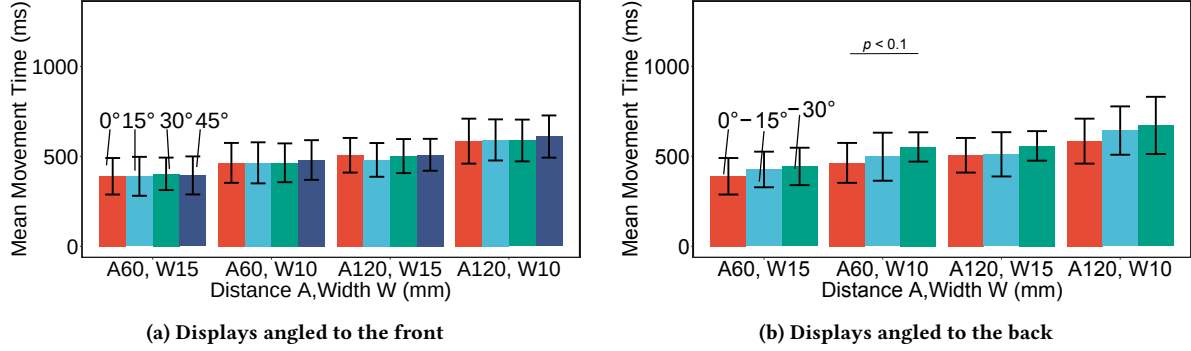


Figure 4: Average tapping time for displays angled in the depth direction.

Therefore, this study proposes an expanded model that considers this situation for prediction.

4 MODELING OF TAPPING TIME

4.1 Method

Based on the hypothesis from the experiment results, our model replaced W in Fitts' law [2] with W_{eyes} , the width of the target when looking at the display from the operator's line of sight, which is expressed as follows:

$$W_{\text{eyes}} = W \cdot \sin(\theta_{\text{eyes}} + \theta), \quad (2)$$

where θ denotes the angle between the display surface and the horizontal plane, and θ_{eyes} denotes the angle between the operator's line of sight and the horizontal plane (Figure 6). In calculating θ_{eyes} ,

we used the average heights from the seat surface to the eyes and elbows in the sitting posture from a database [10] as the eye height from the display varies among individuals. Based on the database, $\theta_{\text{eyes}} = 64.6^\circ$ was adopted in this study. Using ID and W_{eyes} , the tapping task difficulty on an angled display can be expressed as follows:

$$ID_{\text{eyes}} = \log_2\left(\frac{A}{W_{\text{eyes}}} + 1\right). \quad (3)$$

4.2 Results

To verify the predicting ability of ID_{eyes} , a regression analysis was conducted using ID_{eyes} as explanatory variables. Figure 7 presents the results. For displays angled to the front, adjusted $R^2 = 0.97$, indicating that ID_{eyes} has an almost equal or better prediction accuracy than ID . For displays angled to the back, adjusted $R^2 = 0.95$,

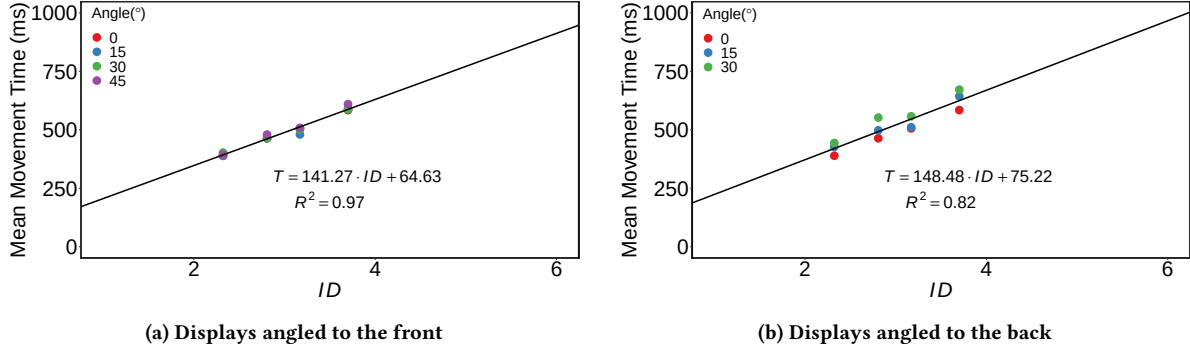


Figure 5: Regression analysis result using ID for displays angled in the depth direction.

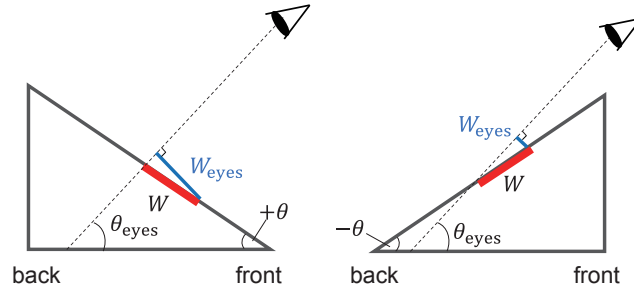


Figure 6: Width of the target when looking from the operator's line of sight.

indicating that ID_{eyes} achieved an improved prediction accuracy compared with ID . These results indicated that the ID_{eyes} effectively modeled tapping time on angled displays.

5 VALIDATION OF THE MODEL FOR MULTIPLE-ANGLED DISPLAYS

5.1 Experiment Setting

An additional experiment was conducted to confirm whether ID_{eyes} applies to various multiple-angled displays, as introduced in Section 2. The environment and method were also the same as in Section 3. The experiment was conducted for the 18 participants described in Section 3. The finger trajectories were also captured using a motion capture camera (Optitrack, V120: Trio) for eight out of the 18 participants. The distance between the motion capture camera and the display edge was 1 m.

5.2 Conditions

Table 2 presents the additional experiment conditions. The target width W and the distance A were set to the same conditions as those in Section 3. The angles of the display to the horizontal were set to 10 conditions; all of them were applied for both convex and concave configurations (Figures 8 and 9). The experiment was conducted under a total of 80 conditions combining these conditions.

5.3 Results and Discussion

Figure 10 presents the results of the regression analysis. The results indicated that adjusted $R^2 = 0.85$ in convex configurations, suggesting that ID_{eyes} applies to various multiple-angled displays. However, in the figure, several conditions with similar ID_{eyes} appeared to be distributed along the vertical axis, and there was variation in the tapping time. This was presumably due to the obstruction of the display height. Because the participants' fingers needed to pass over the highest point of the display in the convex configuration, the travel distance of the finger increased as the display height increased.

For concave configurations, the results indicated that adjusted $R^2 = 0.74$, suggesting that ID_{eyes} is not applicable to various multiple-angled displays. Figure 11 presents the average tapping time for the combinations of 0° and angled to the front. The error bars represent the standard deviations. The tapping time appeared to increase as the angle increases. For the angles of 75° and 90° , tapping time was longer for $A = 60$ than for $A = 120$. Participants also moved their fingers more slowly when the distance A was shorter. The tapping time was presumably increased due to the tapping difficulty. From these viewpoints, we considered expanding ID_{eyes} and modeling the incremental tapping time.

5.4 Expansion of the model

To consider height obstruction in the convex configuration, we modeled the increments in the finger travel distance. As presented in Figure 12, the shortest distance from the start to the target and

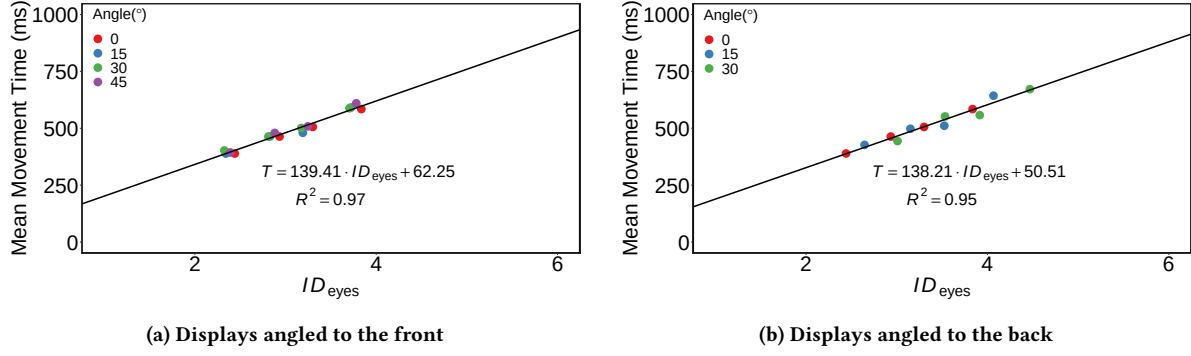
Figure 7: Regression analysis result using ID_{eyes} for displays angled in the depth direction.

Table 2: Conditions for Multiple-angled Displays

Width W (mm)	Distance A (mm)	Angle ($^{\circ}$)	Configuration
10	60	15 & -15, 30 & -30	Convex
15	120	0 & 15, 0 & 30, 0 & 45, 0 & 60, 0 & 75, 0 & 90 0 & -15, 0 & -30	Concave

(a) Angled to the front and back (b) 0° and angled to the front (c) 0° and angled to the back

Figure 8: A set of multiple-angled displays (convex configuration).

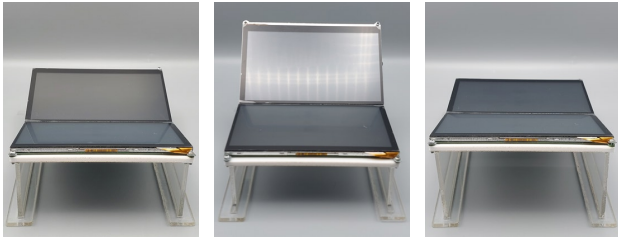
(a) Angled to the front and back (b) 0° and angled to the front (c) 0° and angled to the back

Figure 9: A set of multiple-angled displays (concave configuration).

the distance from the highest point of the display to the line segment connecting the start and the target were defined as A' and h , respectively. Figure 13 presents finger trajectories for the angle of 30° and -30° ($A = 60$, $W = 10$) in the convex configuration. The

red line in the figure represents the average highest point of finger trajectories. The results showed that the highest point of arrival of the finger when tapping the target can be approximated by about 2.1 times the height h . Based on this, the distance that considers the effect of display height obstruction is expressed as follows:

$$A_h = A \cdot \sqrt{1 + 3.84 \cdot \sin^2 \theta}. \quad (4)$$

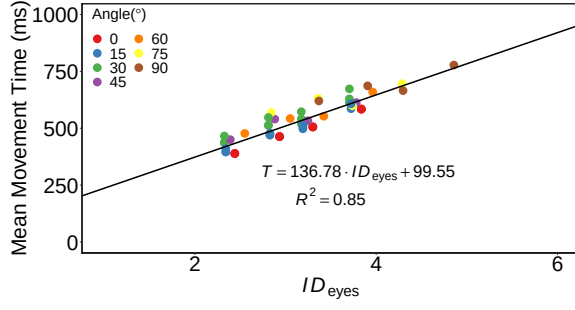
Note that θ becomes $\frac{\theta}{2}$ when either display is 0° . Our expanded model replaced the distance A in Fitts' law [11] with A_h , which considers the obstruction by the display height. Therefore, ID_{h_eyes} , which is an extension of ID_{eyes} , is expressed as follows:

$$ID_{h_eyes} = \log_2 \left(\frac{A_h}{W_{eyes}} + 1 \right). \quad (5)$$

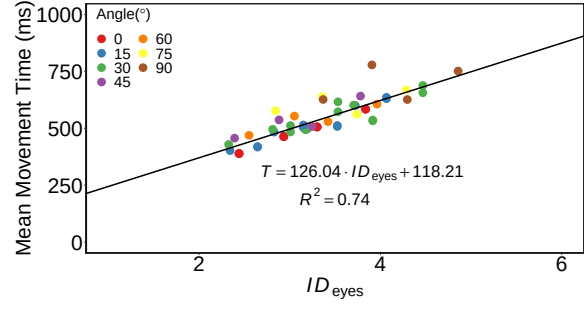
Figure 14 presents the result of a regression analysis that used ID_{h_eyes} as an explanatory variable. The result indicated that adjusted $R^2 = 0.93$, suggesting that ID_{h_eyes} has improved accuracy compared with ID_{eyes} .

To model the tapping difficulty in the concave configuration, we observed participants' finger movement during the tasks. Most participants tried to tap the target with their finger pad as much as possible. The smaller the space available for moving the hand, the more carefully they tapped the target so that their hand did not touch the displays. Figure 15 presents the average finger velocity with respect to display angles in the concave configuration. Because the length of A' , the shortest distance from the start to the target (Figure 16), depends on the display angles, we assumed that the tapping time increases exponentially with the length of A' . The exponent of ratio between the length from the index finger to the wrist and A' is expressed as follows:

$$e^{\frac{\text{hand_size}}{A' \cdot \cos \theta}}, \quad (6)$$



(a) Convex configuration



(b) Concave configuration

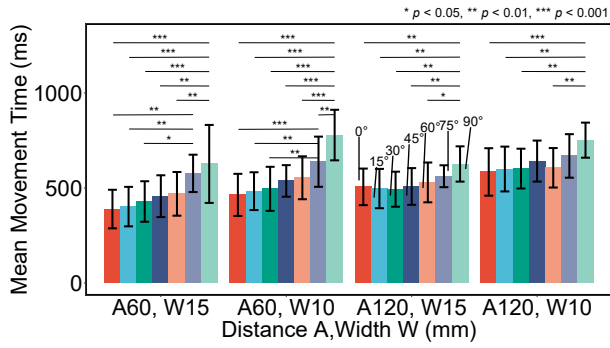
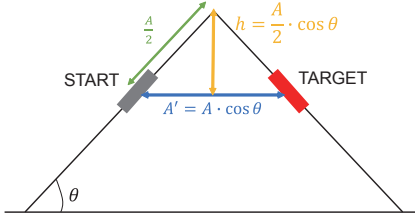
Figure 10: Regression analysis result using ID_{eyes} for multiple-angled displays.Figure 11: Average tapping time for the display combinations of 0° and angled to the front (concave configuration).

Figure 12: Display height.

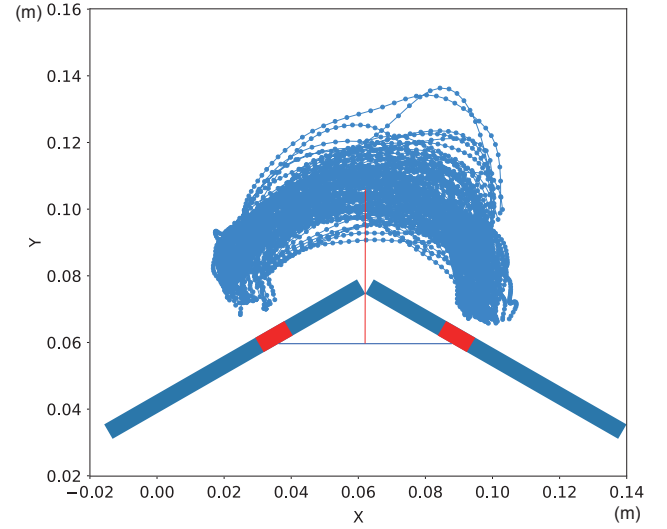
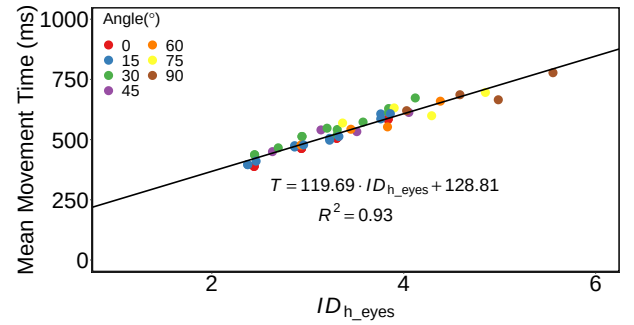
where e is Euler's number and $hand_size$ is 179.5 mm, the average length from the index finger to the wrist in the database [10]. The model of tapping difficulty due to space limitation is expressed as follows:

$$D_{space_lim} = e^{\frac{hand_size}{A \cdot \cos \theta}} \cdot \sin \theta, \quad (7)$$

where $\sin \theta$ is applied so that the incremental tapping difficulty is zero under the condition that the display angle is 0° . The result of a regression analysis that used ID_{eyes} and D_{space_lim} as explanatory variables is as follows:

$$T = 115.75 \cdot ID_{eyes} + 2.72 \cdot D_{space_lim} + 117.28, \quad (8)$$

where adjusted $R^2 = 0.93$, suggesting that ID_{eyes} and D_{space_lim} have improved accuracy compared with ID_{eyes} only. Our expanded

Figure 13: Finger trajectories of 30° and -30° in the convex configuration.Figure 14: Regression analysis result using ID_{h_eyes} .

models predicted the tapping time on various multiple-angled displays both in convex and concave configurations with high accuracy.

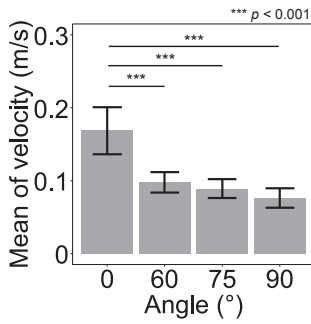


Figure 15: Average finger velocity in the concave configuration.

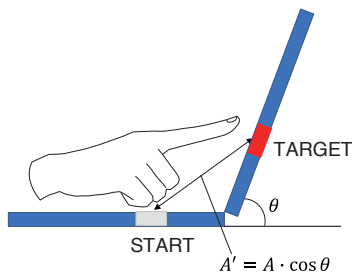


Figure 16: The shortest distance A' in the concave configuration.

6 CONCLUSION AND FUTURE WORK

In this study, we investigated the tapping time of a display angled in the depth direction and expanded Fitts' law, considering the operator's line of sight and hand motion. The results indicate that our expanded models could predict the tapping time for various angled displays. Although our models have high prediction accuracy, consideration of hand motion is still observation-based. In future research, we intend to investigate whether our model can effectively predict tapping time in larger multiple display environments. The proposed future studies will improve the operability of a shape-changing device by optimizing its user interface using our model.

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